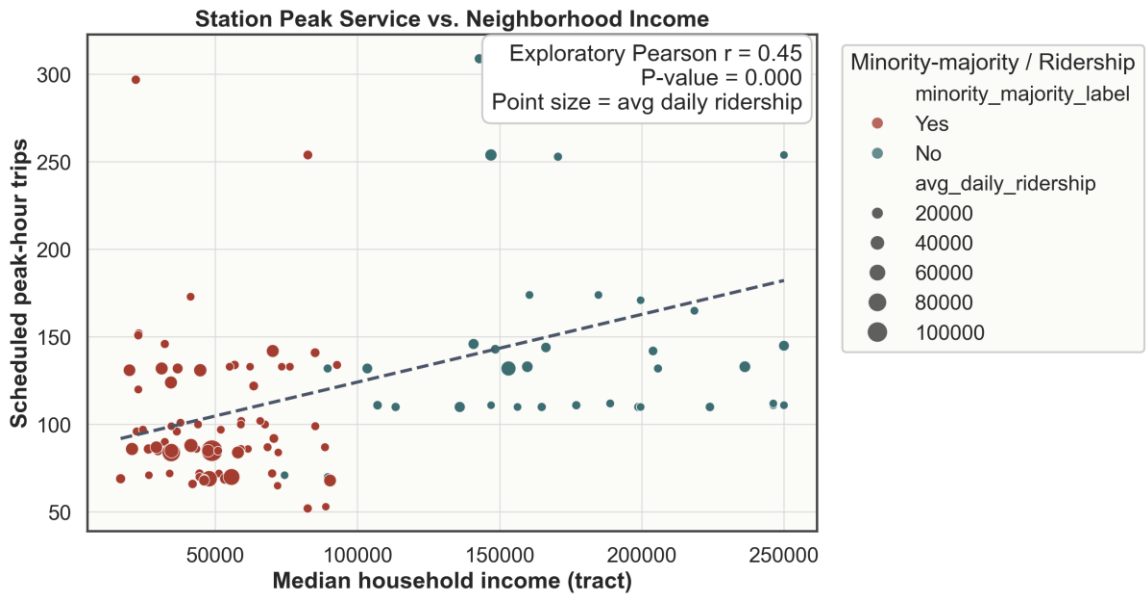
NYC Subway Equity Analysis: Visualization Report

This report summarizes the five key visualizations developed for the NYC Subway Equity Project. These charts transform statistical data into intuitive evidence, illustrating the relationship between neighborhood demographics (income and race) and the quality of transit service.

1. Station Peak Service vs. Neighborhood Income

Filename: h1_income_vs_peak_service.png



What this graph shows

This scatter plot examines whether richer or poorer neighborhoods receive more scheduled subway service during peak commute hours. Each dot represents a station area.

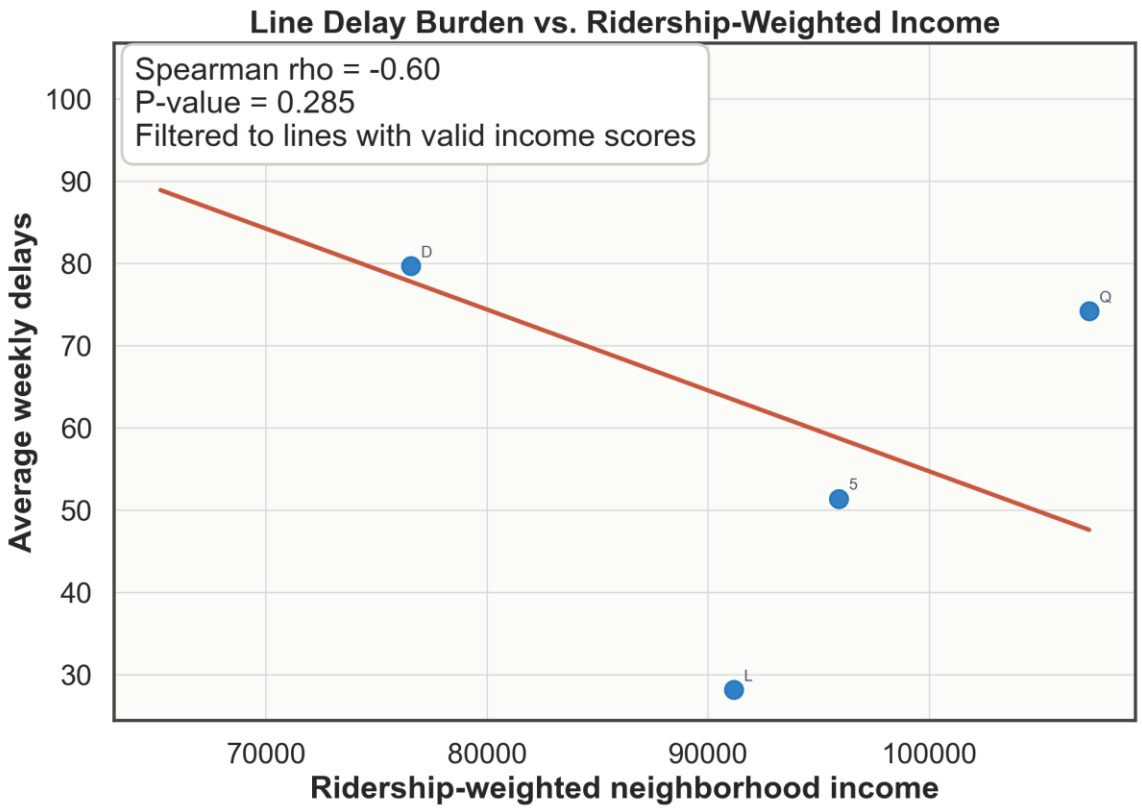
- X-axis: Median household income of the surrounding Census tract.
- Y-axis: Number of scheduled peak-hour trips.
- Dot size: Average daily ridership.
- Dot color: Minority-majority (Brown) vs. non-minority-majority (Teal) tracts.
- Dashed line: The overall trend line (regression).

Main Takeaway & Project Support

The trend line slopes upward, suggesting higher-income neighborhoods generally receive more service. This supports Hypothesis 1 (H1), providing a visual equity comparison between wealth and service frequency.

2. Line Delay Burden vs. Ridership-Weighted Income

Filename: h2_delay_vs_income.png



What this graph shows

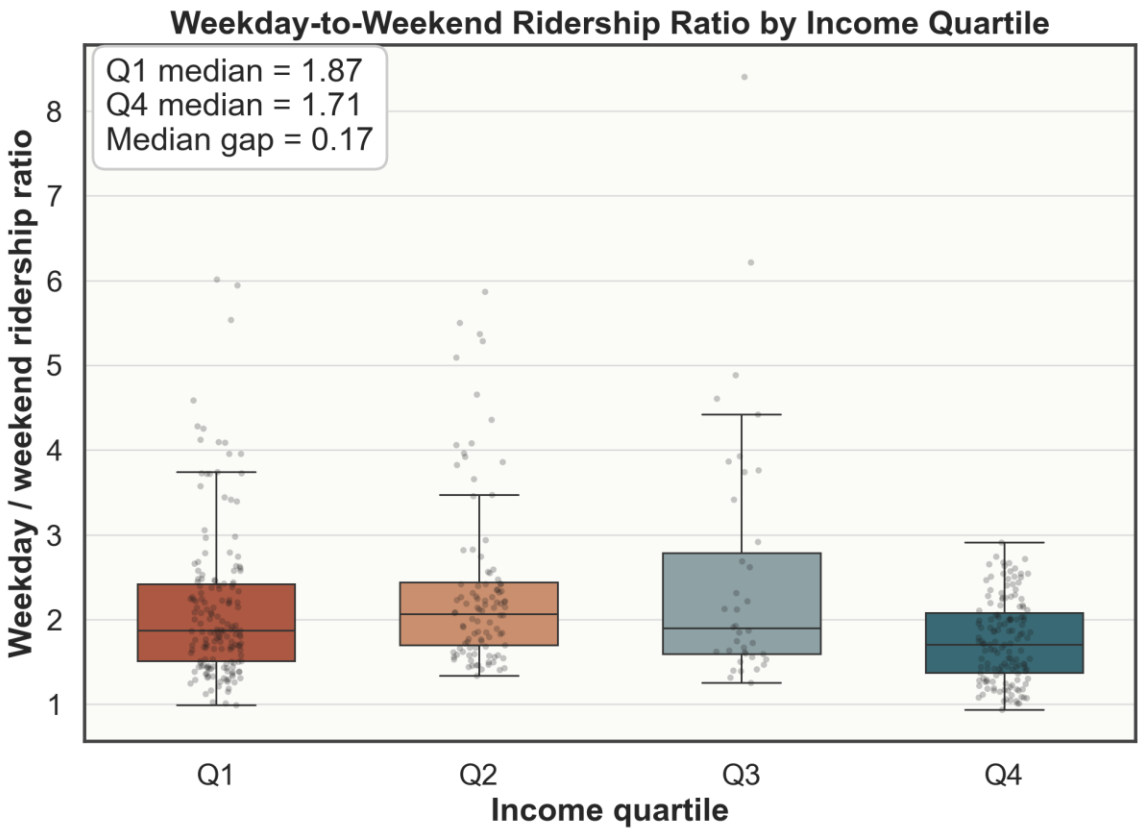
Each point represents a subway line. The x-axis shows the weighted income of the line's ridership, while the y-axis shows average weekly delays.

Main Takeaway & Project Support

This graph checks if lines serving lower-income neighborhoods have more delays. It moves the analysis from 'service frequency' to 'reliability.' It supports H2 by showing how delay burdens may fall disproportionately on lower-income-serving lines.

3. Weekday-to-Weekend Ridership Ratio by Income Quartile

Filename: h3_weekday_weekend_ratio.png



What this graph shows

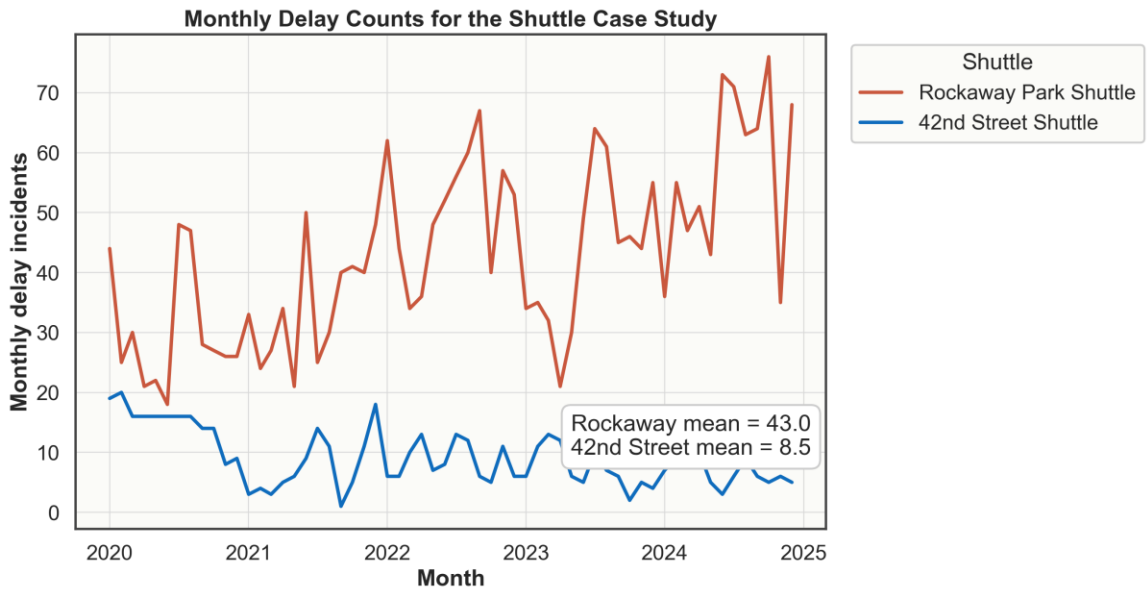
This box plot groups stations into income quartiles (Q1 = lowest, Q4 = highest). The y-axis shows the ratio of weekday to weekend ridership.

Main Takeaway & Project Support

Lower-income stations (Q1) show a higher median ratio (1.87) compared to higher-income stations (Q4, 1.71). This suggests low-income riders rely more on the subway for weekday work commutes, supporting H3 regarding differences in transit dependency.

4. Monthly Delay Counts: Shuttle Case Study

Filename: h4_shuttle_monthly_delays.png



What this graph shows

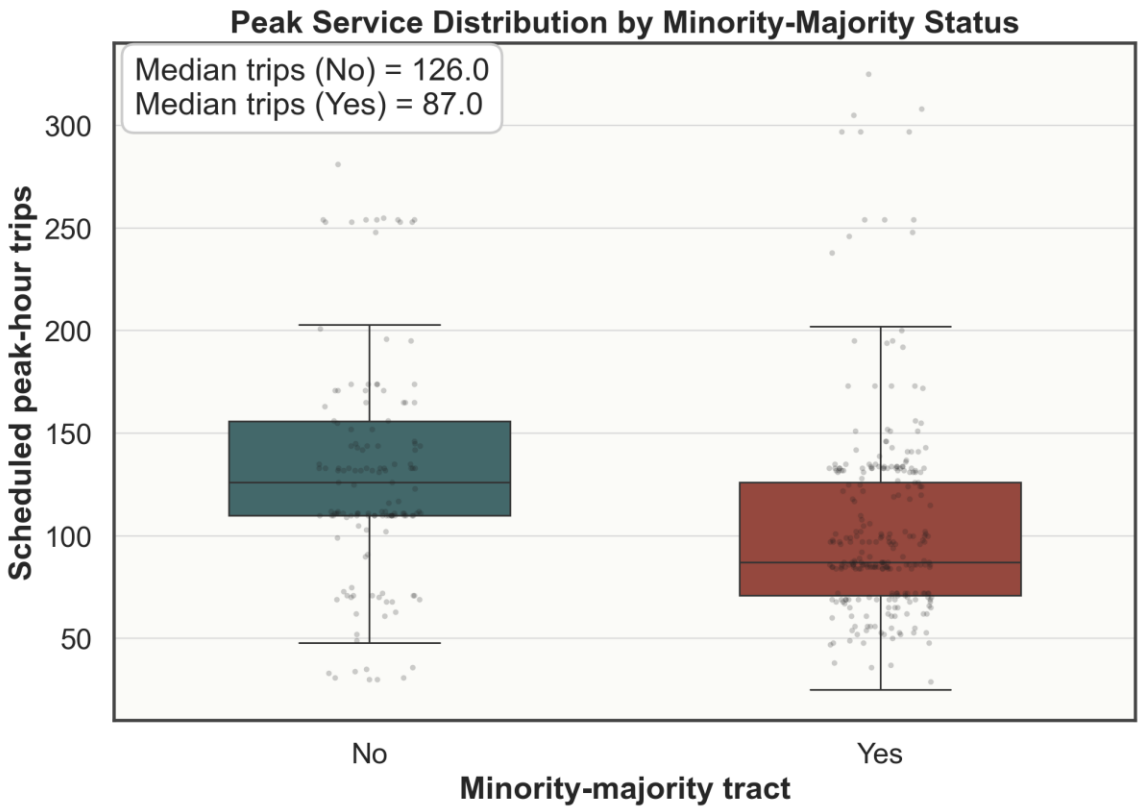
A direct time-series comparison between the Rockaway Park Shuttle (Orange) and the 42nd Street Shuttle (Blue) from 2020 to 2024.

Main Takeaway & Project Support

The Rockaway shuttle experiences far more delays consistently (Mean 43.0 vs 8.5). This supports H4 and is a powerful case study for geographic inequity, as both are shuttle services run by the same agency.

5. Peak Service Distribution by Minority-Majority Status

Filename: equity_overview_minority_majority_service.png



What this graph shows

This compares scheduled peak-hour trips between minority-majority tracts and non-minority-majority tracts using grouped distributions.

Easy Meaning & Project Support

This graph asks if station areas in minority-majority neighborhoods get different service levels regardless of income. The median for minority-majority tracts (87.5) is significantly lower than for others (132.0), supporting the broader equity framing of the project.

Conclusion: How the set works together

Together, these graphs cover the system from multiple angles: frequency (01), reliability (02), behavioral usage (03), a specific case study (04), and demographic equity (05). This multi-layered approach provides robust evidence for the project's claims of transit inequity.